

Example Code W6

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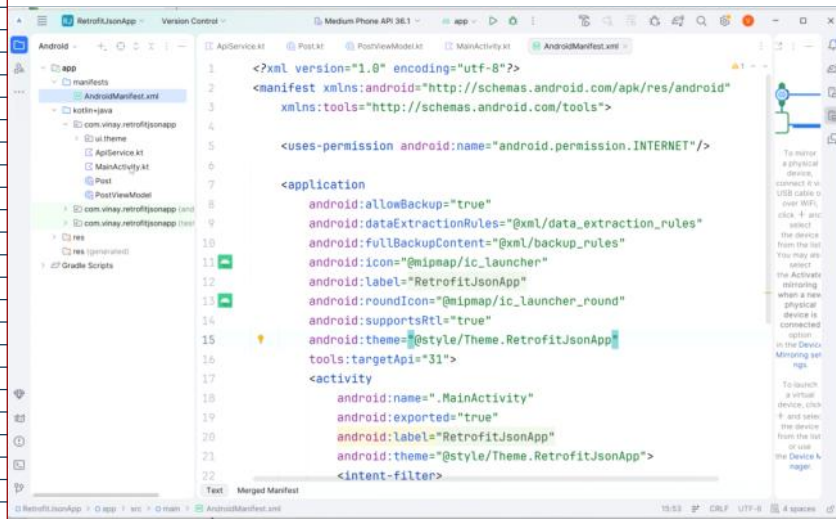


chaptersix



RetrofitJsonApp

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MainActivity.kt

```

1 package com.vinay.retrofitjsonapp
2
3 > import ...
30
1 Usage
31 class MainActivity : ComponentActivity() {
32     override fun onCreate(savedInstanceState: Bundle?) {
33         super.onCreate(savedInstanceState)
34         enableEdgeToEdge()
35         setContent {
36             RetrofitJsonAppTheme {
37                 PostScreen()
38             }
39         }
40     }
41 }
42
1 Usage
43 @OptIn(markerClass = ExperimentalMaterial3Api::class)
44 @Composable
45 fun PostScreen(postViewModel: PostViewModel = viewModel()) {
46     val posts by postViewModel.posts.collectAsState()
47
48     Scaffold(
49         topBar = {
50             TopAppBar(
51                 title = { Text("Posts") }
52             )
53         }
54     ) { paddingValues ->
55         LazyColumn(modifier = Modifier
56             .padding(paddingValues)
57             .padding(all = 16.dp)
58         ) {
59             items(posts) { post ->
60                 PostItem(post)
61             }
62         }
63     }
64 }
65
1 Usage
66 @Composable
67 fun PostItem(post: Post) {
68     Card(
69         modifier = Modifier
70             .fillMaxWidth()
71             .padding(vertical = 8.dp)
72     ) {
73         Column(modifier = Modifier.padding(all = 16.dp)) {
74             Text(
75                 text = post.title,
76                 style = MaterialTheme.typography.titleMedium
77             )
78             Spacer(modifier = Modifier.height(4.dp))
79             Text(
80                 text = post.body,
81                 style = MaterialTheme.typography.bodySmall
82             )
83         }
84     }
85 }
86
87
88

```

Post.kt x

```
1 package com.vinay.retrofitjsonapp
2
3 data class Post(
4     val userId: Int,
5     val id: Int,
6     val title: String,
7     val body: String
8 )
9
```

Data class

ApiService.kt x

```
1 package com.vinay.retrofitjsonapp
2
3 import ...
4
5
6
7 interface ApiService {
8     @GET("posts")
9     suspend fun getPosts(): List<Post>
10 }
11
12 object RetrofitClient {
13     val api: ApiService by lazy {
14         Retrofit.Builder()
15             .baseUrl("https://jsonplaceholder.typicode.com/")
16             .addConverterFactory(GsonConverterFactory.create())
17             .build()
18             .create(ApiService::class.java)
19     }
20 }
```

Helps access the JSON file

PostViewModel.kt x

```
1 package com.vinay.retrofitjsonapp
2
3 import ...
4
5
6
7
8
9 class PostViewModel : ViewModel() {
10     private val _posts = MutableStateFlow<List<Post>>()
11     val posts: StateFlow<List<Post>> = _posts
12
13     init {
14         viewModelScope.launch {
15             try {
16                 _posts.value = RetrofitClient.api.getPosts()
17             } catch (e: Exception) {
18                 // Handle error
19             }
20         }
21     }
22 }
```

Carries out ____?



CoroutineJSONApp

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- Prof accidentally gave us an old example that doesn't use jetpack compose